SERVER:
COMPOSABLE
MEMORY SYSTEMS
(CMS)



Accelerating High-Power Server Design Through Board-Level Electro-Thermal Co-Simulation of VRMs and PDNs

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Outline

Three sections guide the path from board-level simulation challenges to validated simulation results.

01 **PCBA Simulation Challenges**

02 **Electrical – Thermal Co-Simulation Workflow**

03 **Simulation Setup and Result**

PCBA Simulation Challenges

AI servers push the thermal limit

Board-level paths now determine whether the design meets temperature targets.

HIGHER POWER CONSUMPTION FOR AI

+ Power

Modern accelerator boards power components tighter than ever.

MOVING TOWARD FANLESS COOLING

No Airflow

Liquid cooling improves system thermal performance, while local PCBA hot-spots remain a key focus area.

FASTER PRODUCT DEVELOPMENT

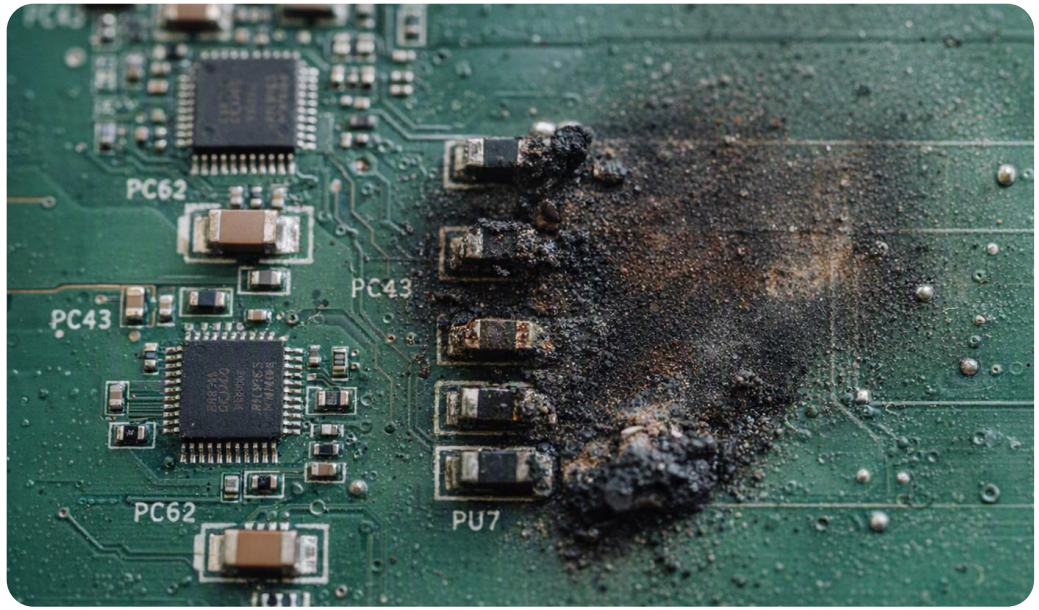
Schedule

Architecture, layout and cooling decisions must be made before all thermal data is in.

Board-level passive components are emerging as a critical thermal design concern.

Long-duration hot spots can escape early validation

Reliability risk increases when the board path is not modeled with enough physics.



AI-generated illustrative image — not actual hardware

01 Local board hot spot

Near power components, board surface temperature exceeds prediction.

02 Late discovery

Most over-temperature scenarios do not affect device function at time zero.

03 No early warning

Standard passive components have no sensors to flag abnormal temperatures, so degradation goes undetected.

This is why early-stage material property assumptions need a stronger physics basis.

Early simulation runs ahead of complete data

Layout is locked early, but full thermal data arrives only after measurement.

DESIGN TIMELINE

01

Concept

Architecture and cooling strategy decisions begin.

02

Initial simulation

Basic board layout is confirmed (simplified k).

03

Measured data rerun

Vendor data arrives; simulation is re-validated.

04

Layout change request

Layout must change — time and headcount cost.

Late layout change request?

Schedule slips, engineering effort climbs, design risk grows.

→ **Time, headcount, and re-routing cost**

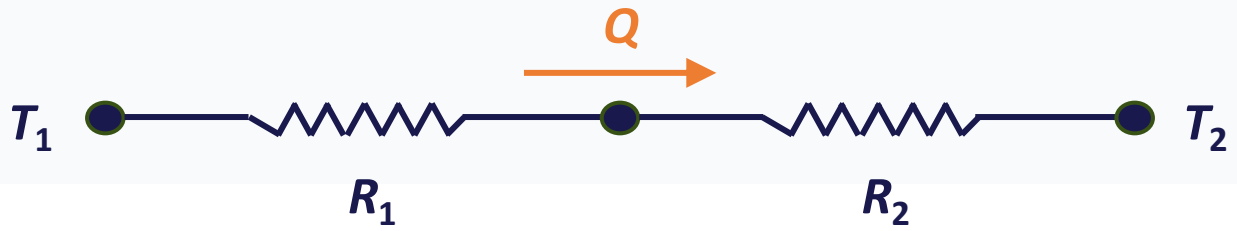
Over-simplified initial k?

Hidden thermal gaps surface only after measurement.

→ **Late discovery, forced rework**

Simulation Setup and Result

Series Model of thermal Conductivity Calculation



01 According to equivalent thermal resistance concept:

$$Q = \frac{T_1 - T_2}{R_{\text{eff}}} = \frac{T_1 - T_2}{R_1 + R_2}$$

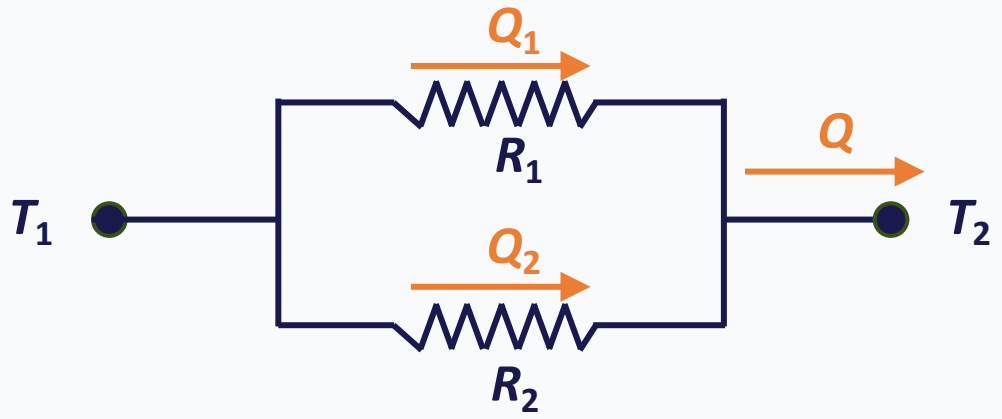
02 Substituting the thermal conduction resistance formula $\mathbf{R} = \frac{\mathbf{L}}{\mathbf{kA}}$ yields:

$$\frac{\mathrm{L}}{k_{eff}\mathrm{A}} = \frac{L_1}{k_1\mathrm{A}} + \frac{L_2}{k_2 A}$$

03 Reformulated in terms of volume fraction:

$$\frac{1}{k_{eff}} = \frac{v_1}{k_1} + \frac{v_2}{k_2} = \sum_{i=1}^{2}\left(\frac{v_{\mathrm{i}}}{k_{\mathrm{i}}}\right)$$

Parallel Model of thermal Conductivity Calculation



01 According to conservation of energy and equivalent thermal resistance concept:

$$Q = Q_1 + Q_2 \qquad \frac{T_1 - T_2}{R_{\text{eff}}} = \frac{T_1 - T_2}{R_1} + \frac{T_1 - T_2}{R_2}$$

02 Substituting the thermal conduction resistance formula $\mathbf{R} = \frac{\mathbf{L}}{\mathbf{kA}}$ yields:

$$k_{eff} \times (A_1 + A_2) = k_1 \times A_1 + k_2 \times A_2$$

$$k_{eff} = k_1 \times v_1 + k_2 \times v_2 = \sum_{i=1}^{2} (k_i \times v_i)$$

Effective Thermal Conductivity Calculation of PCB

- Derived PCB effective thermal conductivity from board structure using series and parallel models, achieving simulation accuracy comparable to models based on actual copper pour percentages

IN-PLANE

$$k_{\text{in}} = \frac{\sum_{i=1}^{n_c} t_{\text{ci}} \times [v_i \times k_{\text{c}} + (1 - v_{\text{i}}) \times k_{\text{D}}] + \left(t - \sum_{i=1}^{n_c} t_{\text{ci}}\right) \times k_{\text{D}}}{t}$$

THROUGH-PLANE

$$k_{\text{thro}} = \frac{t}{\sum_{i=1}^{n_{\text{c}}} t_{\text{ci}} \times \left[\frac{v_{\text{i}}}{k_{\text{c}}} + \frac{(1 - v_{\text{i}})}{k_{\text{D}}}\right] + \frac{\left(t - \sum_{i=1}^{n_{\text{c}}} t_{\text{ci}}\right)}{k_{\text{D}}}}$$

Effective k of conducting layer

k_c = Thermoconductivity of conductor material

k_D = Thermoconductivity of dielectric material

k_{in} = In-plane thermoconductivity

k_{thro} = Through-plane thermoconductivity

n_c = Number of conducting layer

t = PCB total thickness

t_{ci} = Layer thickness

v_i = Volume fraction of conductor material in the layer

PCB Simulation Settings

- **Copper Pour Percentage**
  - Material: FR4 and Copper layer
  - PCB total thickness: 2.34 mm
  - Define numbers, thickness and copper pour percentage of conductor layers

EFFECTIVE THERMAL CONDUCTIVITY

64.2
W/m·K
In-plane direction

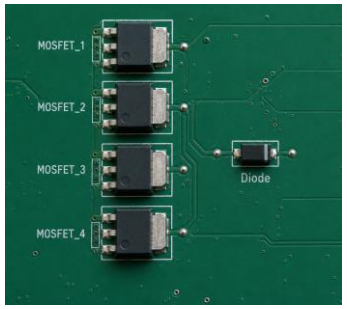
0.36
W/m·K
Through-thickness direction

PCB STACK-UP AND COPPER POUR

Layer	1	2	3	4	5	6	7	8	9	10	11	12
Conducting Layer Thickness (mm)	0.05	0.03	0.03	0.03	0.06	0.06	0.06	0.06	0.03	0.03	0.03	0.05
Copper Pour Percentage (%)	47	91.8	23.3	91.9	81.7	89.8	81.6	82.8	91.9	17	91.7	39.3

Simulation Result of Different PCB Settings

- The effective thermal conductivity model delivers simulation results comparable to those obtained using actual copper pour percentage from the PCB stack-up, while providing a more versatile PCB modeling approach



AI-generated illustrative image — not actual hardware

Components	Simulation Setting	Deviation Between Simulation Result and Test Result
Diode	Copper pour percentage	15%
	Effective Thermal Conductivity	15%
MOSFET_1	Copper pour percentage	8%
	Effective Thermal Conductivity	8%
MOSFET_2	Copper pour percentage	8%
	Effective Thermal Conductivity	8%
MOSFET_3	Copper pour percentage	11%
	Effective Thermal Conductivity	11%
MOSFET_4	Copper pour percentage	10%
	Effective Thermal Conductivity	9%

Effective Thermal Conductivity of Diode

- **According to the material composition table, the volume fraction of each material can be calculated as follows:**

Material	Substance	Mass (mg)	Thermal Conductivity (W/m·K)	Density (g/cm³)	Volume Fraction (%)
Molding compound	SiO_2	0.736	1.13	2.2	75%
	Epoxy Resin	0.050	0.2	1.2	
	Phenol Resin	0.041	0.2	1.1	
Leadframe	Alloy 42	0.578	12.5	8.15	15%
	Ag	0.022	415	10.5	
Die	Si	0.025	139	2.33	3%
Solder	Sn	0.029	60	7.3	5%
Wire	Au	0.005	305	19.3	1%

- **Effective thermal conductivity (k_{eff}) of series structure can be calculated as:**

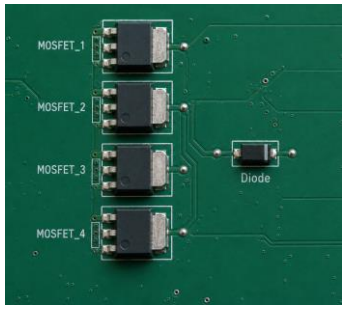
$$\frac{1}{k_{eff}} = \sum_{i=1}^{5} \left(\frac{V_i}{k_i} \right) \longrightarrow k_{eff} = 0.96 \text{ W/m·K}$$

V_i = Volume fraction of material

k_i = thermal conductivity of material

Diode Simulation Result

- Based on the diode material composition and package dimensions, the series model was selected to calculate the diode's effective thermal conductivity
- Since the diode uses an epoxy-based package, the simulation results obtained with the series model were compared with those using the epoxy compound reference properties
- The series model showed better correlation with the test results, reducing the temperature deviation to 3%.



AI-generated illustrative image — not actual hardware

Diode		
Simulation Setting	**k (W/m·K)**	**Deviation Between Simulation Result and Test Result**
Expoxy compound	**0.7 W/m·K**	**10%**
Series Model	**0.96 W/m·K**	**3%**

Effective Thermal Conductivity of MOSFET

- **According to the material composition table, the volume fraction of each material can be calculated as follows:**

Substance	Mass (mg)	Thermal conductivity (W/m·K)	Density (g/cm³)	Volume Fraction (%)
SiO_2	37.959	1.13	2.2	56.5%
Epoxy Resin	5.047	0.2	1.2	13.8%
Cu	70.153	385	8.96	25.7%
Si	1.818	139	2.33	2.6%
Pb	2.380	35	11.34	0.69%
Fe	0.836	80	7.87	1.9%
Sn	1.502	60	7.31	0.67%
Ag	0.228	415	10.49	0.07%
Zn	0.039	116	7.14	0.12%
Carbon black	0.129	5	1.8	0.26%
P	0.021	0.24	1.8	~0.04%
Au	0.044	305	19.3	~0.01%

- **Effective thermal conductivity (k_{eff}) of series structure can be calculated as:**

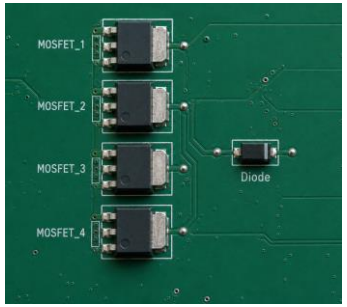
$$k_{eff} = \sum_{i=1}^{12} (k_i \times v_i) \longrightarrow k_{eff} = 106 \text{ W/m·K}$$

V_i = Volume fraction of material

k_i = thermal conductivity of material

MOSFET Simulation Result

- The effective thermal conductivity model is a practical alternative to the 2R model, providing comparable simulation accuracy with only a 4–5% deviation compared to the 2R model.



AI-generated illustrative image — not actual hardware

MOSFET			
Components	**Simulation Setting**	**Deviation Between Simulation Result and Test Result**	**Deviation between 2R model and Effective Thermal Conductivity Results**
MOSFET_1	2R Model	8%	4%
	Effective Thermal Conductivity	12%	
MOSFET_2	2R Model	8%	5%
	Effective Thermal Conductivity	13%	
MOSFET_3	2R Model	11%	4%
	Effective Thermal Conductivity	15%	
MOSFET_4	2R Model	10%	4%
	Effective Thermal Conductivity	14%	

Electrical – Thermal Co-Simulation Workflow

Electrical Loss and Temperature Solved Together

Each solver passes its result to the other, so the PI and thermal answers stay consistent.

TEMPERATURE-AWARE PI SIMULATION

Conductor resistance and VRM loss are evaluated at the board temperature the design actually reaches.

- IR drop resolved at true operating temperature
- Margin confirmed against the real thermal state
- Reliability supported across the full load range

POWER LOSS





TEMPERATURE

LOSS-AWARE THERMAL SIMULATION

Heat sources are applied exactly where the electrical losses are generated on the board.

- Hot spots resolved where the loss occurs
- More detailed PCB model in the thermal simulation
- PDN copper and via I^2R heating included

One coupled loop, one consistent answer — in both the electrical and the thermal domain.

Electrical Thermal Co-Simulation

Three steps in sequence — the thermal result returns to a final PI run, and the workflow ends there.



01 PI SIMULATION

Simplified thermal boundary condition

Board stackup · Current/load data · Layout · BGA/QFN ball map / pin map

02 THERMAL SIMULATION

Simplified model from PI simulation

Components' effective thermal conductivity — used when the vendor 2R model is not yet available in the early design phase

Ambient temperature and airflow from real operating conditions

03 PI SIMULATION · UPDATED

Updated thermal boundary condition from the thermal result

Same stackup, current/load, layout and ball map — now solved with the converged temperature field

Mesh Count Reduction and Modeling Time Savings

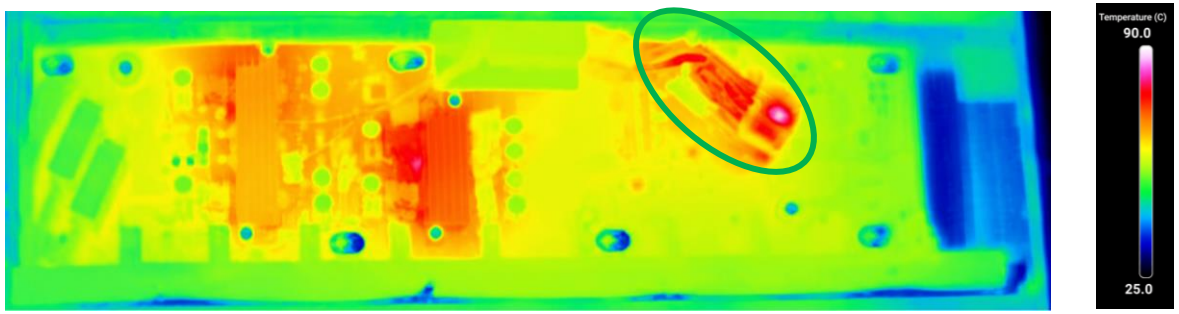
- The thermal simulation in ET cosim uses a simplified model derived from the PI simulation, which effectively reduces both the simulation time and mesh count

Simulation Method	Mesh Number	Calculation Time	Deviation with Test Result
Traditional thermal simulation method	1X	1X	1X
Thermal simulation in ET-cosim process	0.54X (-46%)	0.55X (-45%)	0.9X

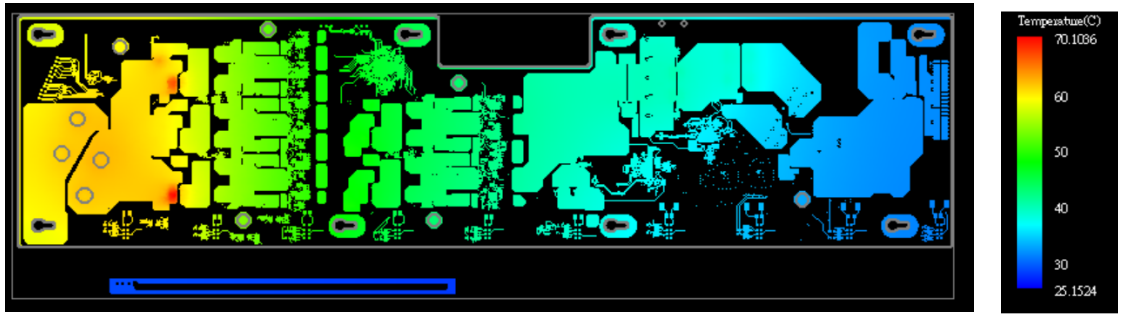
PCB Thermal Profile Comparison

- The ET co-simulation thermal profile matches the IR photo much more closely than the non-ET co-simulation result, in both hotspot location and overall temperature distribution

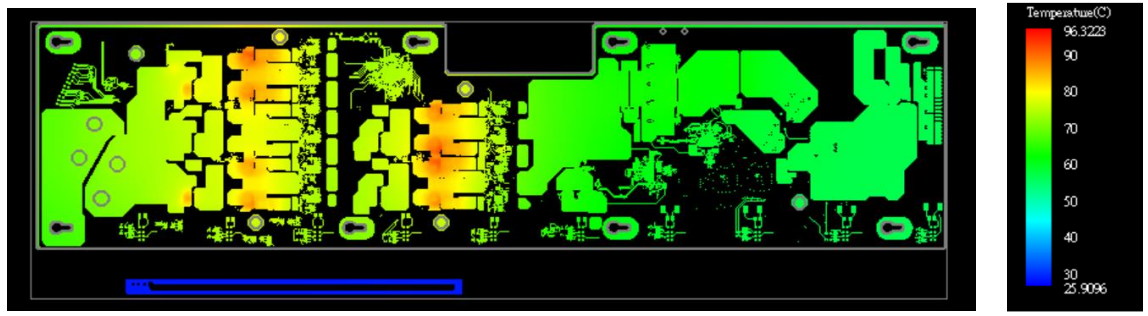
Power Cable



IR Photo



Non ET- Cosim



ET- Cosim

Conclusion

- A physics-based approach improves PCB modeling flexibility by allowing the transformation of layer-defined copper pour percentage into effective thermal conductivity
- Comparable simulation results are obtained when the PCB material is defined using either effective thermal conductivity or copper pour percentage
- A material composition-based approach to determining effective thermal conductivity supports component-level simulation when proprietary vendor thermal data is limited
- The final PI report generated from ET co-simulation reproduces the measured IR thermal profile far more closely than a traditional, non-coupled PI report, matching both hotspot location and overall temperature distribution.
- By using a simplified model derived from the PI simulation, the thermal simulation reduces both mesh count and modeling time, enabling faster design cycle turnover

Call to Action

- Extend the evaluation to different package types and advanced 2.5D/3D packaging to assess applicability and modeling considerations
- Establish an effective thermal conductivity library for different package and component types to support early-stage thermal design evaluation.
- Get involved in the server community:
  - Link: Server - OpenCompute
- Mailing list:
  - Server: OCP-Server@OCP-All.groups.io | Home

Wiwynn Whitepaper

Thank You!

